Project Proposal: AI-Generated Text Detection for Academic Integrity

Hamza Ahmed[1], Muhammad Mouid Ali[1], and Hamza Munim[1]

Fast National University of Computer and Emerging Sciences (NUCES), Islamabad, Pakistan
i221339@nu.edu.pk
i221026@nu.edu.pk
i221130@nu.edu.pk

Abstract. Project Title: AI-Generated Text Detection: A Comparative Analysis of Feature and Transformer-Based Classifiers for Identifying Human vs. AI Essays.

The rapid development of advanced Large Language Models (LLMs) poses a significant threat to academic integrity by enabling the creation of human-quality coursework. Existing AI detection systems often exhibit considerable unreliability when faced with new LLMs and sophisticated evasion tactics. This project addresses the critical need for robust and dependable detection by performing a rigorous comparative analysis of core machine learning architectures and training methodologies.

Our methodology contrasts three distinct approaches to detection:

1. **Supervised Transformer Learning:** Utilizing state-of-the-art architectures like RoBERTa via fine-tuning, with a focus on benchmarking against hybrid models augmented by explicit linguistic features.
2. **Feature-Based Classification:** Implementing traditional classifiers (e.g., SVM, Logistic Regression) using engineered linguistic features, including a novel composite Naturalness Score (based on lexical diversity, readability, and stylistic cues).
3. **One-Class Anomaly Detection:** Exploring One-Class Learning (OCL) models (e.g., One-Class SVM or AutoEncoder) to train exclusively on human text, treating AI output as a detectable anomaly.

The project culminates in a comprehensive discussion on the trade-offs between detection accuracy, interpretability, and most critically, robustness against adversarial attempts to fool the model. This comparative framework will provide vital insights into developing the next generation of resilient AI content detectors for higher education.

1 Introduction and Problem Statement

The fundamental problem addressed is the erosion of trust in educational and journalistic content due to the indistinguishable nature of human- and AI-generated text. The advent of powerful LLMs presents a significant threat to academic integrity by enabling the creation of human-quality coursework. Recent works highlight that while dedicated AI detection classifiers can achieve high accuracy on benchmark datasets (e.g., 99.73% F1-score on HC3), this performance often collapses under real-world conditions involving sophisticated evasion tactics or generalization across diverse domains. The core challenge is building a detection system that achieves high performance while maintaining interpretability and resilience, particularly when facing AI output crafted with **advanced prompting** (i.e., instructed to write "in a way a human would do it"). We aim to specifically address and improve detection performance on this challenging class of "humanized" AI text. [5]

1.1 Project Objectives

The primary objective of this project is to conduct a rigorous **comparative analysis** of different detection architectures and training methodologies across three distinct axes to identify the most robust strategy. These objectives are:

1. **Supervised Feature Engineering:** To implement a comprehensive feature pipeline utilizing linguistic attributes such as **perplexity**, **readability**, **burstiness**, and a composite **Naturalness Score** to train classical machine learning models (e.g., SVM, XGBoost). [8]

2. **Transformer-Based Comparison:** To fine-tune various advanced transformer architectures (primarily **RoBERTa**, and possibly DistilBERT) for binary classification, comparing a **vanilla fine-tuning** approach against a **feature-augmented hybrid** approach. The selection is supported by fundamental work on pre-trained deep bidirectional representations and their linguistic biases. [2] [4] [7] [10]

3. **Methodology Comparison (One-Class Learning):** To explore and evaluate **One-Class Learning (OCL)** models (e.g., One-Class SVM or AutoEncoder) as an alternative training methodology, using exclusively human data and treating AI-generated content as a detectable anomaly. [1]

4. **Performance and Robustness Assessment:** To evaluate all models' performance using standard metrics (**AUROC, Precision, Recall, and F1-score**) and rigorously investigate their **robustness by simulating evasion tactics** and testing specifically against humanized AI-generated text. [6]

2 Theoretical Background and Related Work

The following sections relate our proposed comparative analysis—of supervised feature, transformer, and one-class learning models—to the existing state-of-the-art in Generative AI detection.

2.1 Foundation Models and Architecture

Our project's core deep learning toolset relies on the evolution of the transformer architecture for text classification. The foundation is established by the original **BERT** [2] model, which proved the efficacy of pre-training deep, bidirectional representations. We select **RoBERTa** [4] [9] as our primary architecture because it is a robustly optimized variant of BERT shown to achieve higher performance in classification tasks. Furthermore, research demonstrates that pre-training grants models a crucial **inductive bias** to prioritize robust linguistic patterns over superficial artifacts, a key element for building a resilient detector. The project will also consider **DistilBERT** [7] as a benchmark for deploying computationally efficient models via knowledge distillation.

2.2 Feature Engineering and Comparative Methodologies

Feature-Based and Hybrid Methods We focus on identifying stylistic artifacts absent in machine-generated text. Our approach incorporates classic detection features—such as statistical anomalies (e.g., **Perplexity** and word rank analysis) and structural indicators (e.g., **Repetitiveness, Readability, and Document features**). These methods inform our construction of a novel composite metric, the **Naturalness Score** [3]. The success of this dual approach is validated by recent work showing that integrating explicit linguistic features into a fine-tuned RoBERTa classifier yields state-of-the-art accuracy. The foundational linguistic differences between human and AI text further guide our feature selection process.

Methodology Comparison (One-Class Learning) To address data limitations and the evolving nature of AI output, we benchmark traditional supervised learning against **One-Class Learning (OCL)** [1]. This alternative methodology justifies training solely on verified human essays, treating any AI-generated content—which often lacks human-like irregularity—as a detectable anomaly or outlier.

2.3 Adversarial Challenges and Robustness

The motivation for our rigorous comparative analysis is rooted in the failures of current detection systems under real-world pressure. The threat to academic integrity is pervasive, and standard detectors frequently demonstrate **unreliability** when facing outputs from advanced LLMs or content explicitly modified using **evasion tactics**. The project's central goal—improving performance against humanized AI text—necessitates a comparative framework to identify which combination of architecture and methodology offers the highest level of **robustness and resilience** against these adversarial challenges.

3 Proposed Methodology and Main Functions (Required)

3.1 Datasets and Resources

The core dataset will be a synthesized corpus combining known human writing (e.g., academic samples) with modern AI generations (e.g., essays produced by current LLMs). We will utilize existing corpora for benchmarking and create custom subsets to reflect typical student essay domains.

3.2 Main Functions / Technical Modules

1. **Data Curation and Labeling**: Assemble the Human(0) vs. AI(1) corpus, ensuring balanced labels and diverse prompts across various academic topics.
2. **Feature Engineering Pipeline**: Implement two parallel feature paths:
   - **Statistical Features**: Extraction of Perplexity, Burstiness, and stylistic measures (e.g., Flesch-Kincaid, POS tag distribution).
   - **Deep Embeddings**: Generate document-level vector representations using the output layers of **RoBERTa** and possibly its distilled version, **DistilBERT**.
3. **Model Training and Fine-Tuning**:
   - **Feature Models**: Train classical classifiers (e.g., SVM, XGBoost) on the numerical statistical features to establish a baseline.
   - **Transformer Model**: Fine-tune a pre-trained $\text{RoBERTa}_{\text{BASE}}$ model specifically for the binary classification task.
4. **Adversarial Testing and Evaluation**: Assess the performance using **AU-ROC, Precision, Recall, and F1-score**. Critically, test models against evaded content (e.g., paraphrased AI text) to measure true robustness, a key requirement for next-generation detectors.

4 Expected Results and Discussion Plan

We expect the fine-tuned RoBERTa models incorporating linguistic features to significantly outperform purely classical models. The **F1-score** on the adversarial test set will be the primary metric for concluding which detection method is superior. The final research paper will include detailed discussion on which feature sets provide the best interpretability and most effectively differentiate between human-like diversity and machine-like predictability.

References

1. Corizzo, R., Leal-Arenas, S.: One-class learning for ai-generated essay detection. Applied Sciences **13**(13), 7901 (2023)
2. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). pp. 4171–4186 (2019)
3. Gehrmann, S., Strobelt, H., Rush, A.M.: Gltr: Statistical detection and visualization of generated text. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations. pp. 111–116 (2019)
4. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692 (2019)
5. Lund, B.D., Wang, T.: Ai and academic integrity: Exploring student perceptions and implications for higher education. Journal of the Association for Information Science and Technology **74**(5), 555–566 (2023)
6. Pudasaini, S., Miralles, L., Lillis, D., Salvador, M.L.: Benchmarking AI text detection: Assessing detectors against new datasets, evasion tactics, and enhanced LLMs. In: Proceedings of the 1stWorkshop on GenAI Content Detection (GenAIDetect). pp. 68–77. International Conference on Computational Linguistics, Abu Dhabi, UAE (Jan 2025)
7. Sanh, V., Debut, L., Chaumond, J., Wolf, T.: Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108 (2019)
8. Schaaff, K., Schlippe, T., Mindner, L.: Classification of human- and ai-generated texts for different languages and domains. International Journal of Speech Technology **27**(4), 935–956 (2024)
9. Warstadt, A., Zhang, Y., Li, H., Liu, H., Bowman, S.R.: Learning which features matter: Roberta acquires a preference for linguistic generalizations (eventually). CoRR **abs/2010.05358** (2020), `https://arxiv.org/abs/2010.05358`
10. Yadagiri, A., Shree, L., Parween, S., Raj, A., Maurya, S., Pakray, P.: Detecting ai-generated text with pre-trained models using linguistic features. In: Proceedings of the 21st International Conference on Natural Language Processing (ICON). pp. 188–196. NLP Association of India (NLPAI) (2024)